

STATISTICAL STRUCTURED PREDICTION

Part I: [José Miguel Benedí](#) **e-mail:** jmbenedi@prhlt.upv.es

[Joan Andreu Sánchez](#) **e-mail:** jandreu@prhlt.upv.es

Part II: [Enrique Vidal](#) **e-mail:** evidal@prhlt.upv.es



UNIVERSITAT
POLITÀCNICA
DE VALÈNCIA



Syllabus

Part I: Statistical Structured Prediction for structured output data

1. Introduction
2. Models for Statistical Structured Prediction
3. Making Predictions: Decoding and Inference
4. Model Parameter Estimation

Part II: Fundamentals of Multimodal Interaction in Pattern Recognition

Bibliography of Part I

Statistical Models for Natural Language Processing

- Christopher D. Manning and Hinrich Schtze: **Foundations of Statistical Natural Language Processing**. The MIT Press, 1999.
- Daniel Jurafsky and James H. Martin: **Speech and Language Processing**. Prentice Hall, 2007 (2ed), 2009.

Pattern Recognition and Machine Learning

- Richard O. Duda, Peter E. Hart and David G. Stork: **Pattern Classification** (2nd ed.). Wiley Interscience, 2000.
- Christopher M. Bishop: **Pattern Recognition and Machine Learning**. Springer, 2006.
- Kevin P. Murphy: **Machine Learning: A Probabilistic Perspective**. The MIT Press, 2012.

Formal Languages Theory

- M.A.Harrison: **Introduction to Formal Languages Theory**. Addison Wesley, 1978.
- J.E.Hopcroft and Jeffrey D. Ullman: **Introduction to Automata Theory, Languages and Computation**. Addison Wesley, 1979.

Probability Theory

- Morris H. Degroot: **Probability and Statistics**. Addison Wesley, 1986.

Information Theory

- Thomas M. Cover and Joy A. Thomas: **Elements of Information Theory**. John Wiley and Sons, 1991.

Natural Language Processing

- James Allen: **Natural Language Understanding**. The Benjamin/Cummings Publishing Company, Inc., 1995.

STATISTICAL STRUCTURED PREDICTION

Lectures

Monday 15:00 - 18:00 (10 sessions of 3 hours)

Part I **J.M. Benedí** November 07 first session

J.A. Sánchez December 5 first session

Part II **E. Vidal** January 9 first session

Assessment

- 3 Collections of selected exercises (theoretical and/or practical)
- 1 Multiple-choice test (Monday, February 13, 2023)

	delivery	deadline
Q2	30 - 12 - 2022	13 - 01 - 2023

Tutoring

Tutoring available by previous appointment

4. Model Parameter Estimation

4.1. Introduction

4.2. Unsupervised Learning: Generative Models

- HMMs: *Forward-Backward* algorithm
- PCFGs: *Inside-outside* algorithm
- PCFGs: estimation from a subset of derivations

4.3. Supervised Learning: Generative Models

- Discriminative Training

4.4. Supervised Learning: Discriminative Models

- Parameter estimation in CRFs
- Learning structured predictors: Structured Perceptron

4.5. Advanced Issues in Model Parameter Estimation

- On-Line learning
- Active learning

Let $\mathcal{D}$ be a training corpus, and given a statistical model defined by the parameter θ , the problem to learn θ can be formulated as:

$$\hat{\theta} = \arg \max_{\theta} L_{\theta}(\mathcal{D})$$

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1. Optimization methods:

- Analytical optimization
- Constrained optimization: Lagrange multipliers
- Gradient descent
- Expectation-maximization algorithm
- Growth transformations (Baum-Eagon and extended Baum-Eagon methods)

2. Learning strategies:

- Given a fully annotated training set $\mathcal{D} = \{(x^{(n)}, y^{(n)})\}_{n=1}^N$ Supervised learning
- Given a training set only with inputs $\mathcal{D} = \{x^{(n)}\}_{n=1}^N$ Unsupervised learning

Other learning strategies

- **Semi-supervised learning** makes use of a small amount of labeled data along with a large amount of unlabeled data.
- **Reinforcement Learning** allows models to be learnt by maximizing their performance.
A reward feedback (reinforcement signal) is required for the model to learn the ideal behaviour within a specific context.
- **On-Line learning**
- **Active learning**

3. Objective functions

➤ Maximum likelihood (generative) estimation

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➤ Unsupervised learning

$$L_{\theta}(\mathcal{D}) = \prod_{n=1}^N P_{\theta}(x^{(n)}) = \prod_{n=1}^N \sum_{y \in \theta(x^{(n)})} P_{\theta}(x^{(n)}, y) = \prod_{n=1}^N \sum_{y \in \theta(x^{(n)})} P_{\theta}(x^{(n)}|y) \cdot P_{\theta}(y)$$

- Maximum conditional likelihood (discriminative) estimation (Supervised learning)

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- Maximum mutual information (discriminative) estimation (Supervised learning)

$$L_{\theta}(\mathcal{D}) = \sum_{n=1}^N \log \left[\frac{P_{\theta}^{\alpha}(x^{(n)} | y^{(n)}) \cdot P_{\theta}^{\alpha}(y^{(n)})}{\sum_{y \in \theta(x^{(n)})} P_{\theta}^{\alpha}(x^{(n)} | y) \cdot P_{\theta}^{\alpha}(y)} \right]$$

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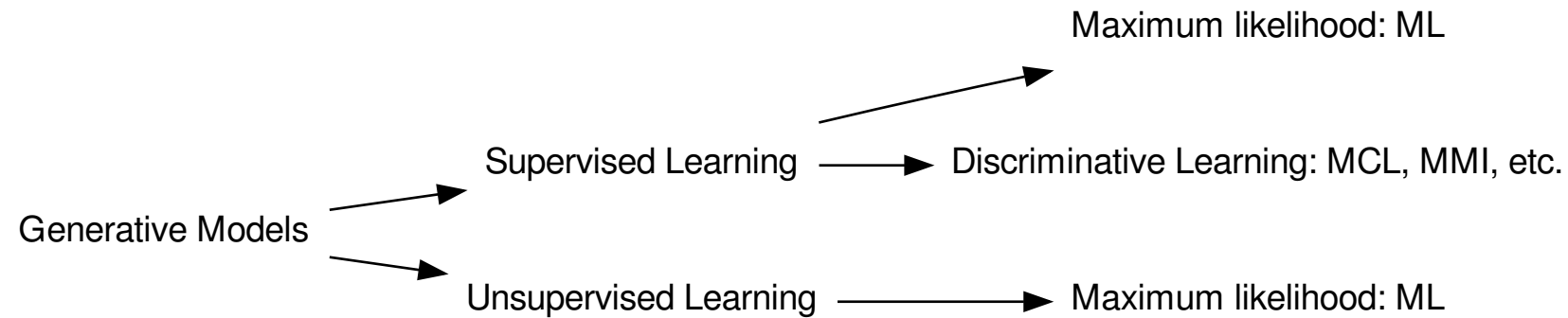
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- Minimum classification error, ...

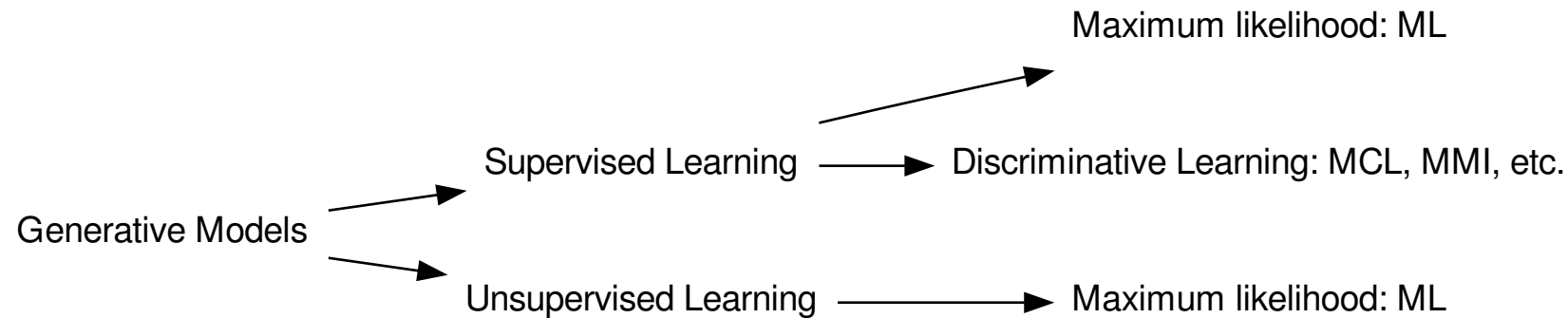
MODEL PARAMETER ESTIMATION

Discriminative Models → Supervised Learning → Discriminative Learning: MCL, MMI, etc.



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Discriminative Models → Supervised Learning → Discriminative Learning: MCL, MMI, etc.



- L.R.Bahl, P.F.Brown, P.V.De Souza, R.L.Mercer (1986) **Maximum mutual information estimation of hidden Markov model parameters for speech recognition**. In Proceedings of IEEE (ICASSP'86):49-52.
- L.E.Baum, G.Sell (1968) **Growth transformation for functions on manifolds**. Pacific Journal of Mathematics 27(2):211-227.
- A.P.Dempster, N.M.Laird, D.B.Rubin (1977) **Maximum likelihood from incomplete data via the EM algorithm (with discussion)**. Journal of the Royal Statistical Society B39:1-38.
- P.S.Gopalakrishnan, D.Kanevsky, A.Nadas, D.Nahamoo (1991) **An inequality for rational functions with applications to some statistical estimation problems**. IEEE Transactions on Information Theory 37(1):107-113.
- T.Jebara, A.Pentland (1998) **Maximum conditional likelihood via bound maximization and the CEM algorithm**. In: Proceedings of Advances in Neural Information Processing Systems, vol.11.

Objective function: **(Log -) Maximum Likelihood estimation**

$$\log L_{\theta}(\mathcal{D}) = \log \prod_{x \in \mathcal{D}} P_{\theta}(x) = \log \prod_{x \in \mathcal{D}} \sum_{y \in \theta(x)} P_{\theta}(x, y) = \sum_{x \in \mathcal{D}} \log P_{\theta}(x)$$

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Optimization method: **Growth Transformations**

Baum and Eagon Theorem [Baum and Sell, 68] [Gopalakrishnan et al., 91]

Let $L(\theta)$ be a homogeneous polynomial with non-negative coefficients. Let $\theta = \{\theta_{ij}\}$ be a point in the domain $\Theta = \{\theta_{ij} \mid \theta_{ij} \geq 0; \sum_{j=1}^{q_i} \theta_{ij} = 1, i = 1, \dots, p; \}$, and let $Q(\theta)$ be a close transformation in Θ , that is defined as:

$$Q(\theta_{ij}^{(w)}) = \theta_{ij}^{(w+1)} = \frac{\theta_{ij}^{(w)} \left(\frac{\partial L}{\partial \theta_{ij}} \right)_{\theta_{ij}=\theta_{ij}^{(w)}}}{\sum_{k=1}^{q_i} \theta_{ik}^{(w)} \left(\frac{\partial L}{\partial \theta_{ik}} \right)_{\theta_{ik}=\theta_{ik}^{(w)}}}$$

with the denominator different from zero. Then, $L(Q(\theta)) > L(\theta)$ except if $Q(\theta) = \theta$

```
input  $L(\theta)$ 
 $\theta$  = initial values
repeat
    compute  $Q(\theta)$  using  $L(\theta)$ 
     $\theta = Q(\theta)$ 
until convergence
output  $\theta$ 
```

```

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 $\theta$  = initial values
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until convergence
output  $\theta$ 
    
```

$$\bar{\theta}_{ij} = \frac{\theta_{ij} \frac{\partial \log P_{\theta}(\mathcal{D})}{\partial \theta_{ij}}}{\sum_{k=1}^{q_i} \theta_{ik} \frac{\partial \log P_{\theta}(\mathcal{D})}{\partial \theta_{ik}}} = \frac{\theta_{ij} \frac{\partial \log \prod_{x \in \mathcal{D}} P_{\theta}(x)}{\partial \theta_{ij}}}{\sum_{k=1}^{q_i} \theta_{ik} \frac{\partial \log \prod_{x \in \mathcal{D}} P_{\theta}(x)}{\partial \theta_{ik}}}$$

$$\frac{\partial \sum_{x \in \mathcal{D}} \log P_{\theta}(x)}{\partial \theta_{ij}} = \frac{\sum_{x \in \mathcal{D}} \frac{\partial \log P_{\theta}(x)}{\partial \theta_{ij}}}{\partial \theta_{ij}} = \sum_{x \in \mathcal{D}} \frac{1}{P_{\theta}(x)} \frac{\partial P_{\theta}(x)}{\partial \theta_{ij}}$$

$$\bar{\theta}_{ij} = \frac{\sum_{x \in \mathcal{D}} \frac{\theta_{ij}}{P_{\theta}(x)} \frac{\partial P_{\theta}(x)}{\partial \theta_{ij}}}{\sum_{x \in \mathcal{D}} \frac{1}{P_{\theta}(x)} \sum_{k=1}^{q_i} \theta_{ik} \frac{\partial P_{\theta}(x)}{\partial \theta_{ik}}} \quad (1)$$

HMMs: FORWARD-BACKWARD ALGORITHM

Given a HMM $(\mathcal{Q}, \Sigma, \theta)$, where $\mathcal{Q}$ is a set of states, Σ is a set of output symbols, and θ is a vector of parameters that contains two types of parameters:

$q(u|v) : u, v \in \mathcal{Q}$ is the **transition probability** from state v to state u .

$e(b|u) : u \in \mathcal{Q}, b \in \Sigma$ is the **probability of emitting symbol** b from state u .

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For the parameter $q(u|v)$, we can express $P_\theta(x)$ as:

$$\sum_t \sum_{\Psi_t(v, u, x_t)} \underbrace{p(x_1^{t-1}, y_{t-1} = v)}_{\alpha_{t-1}(v)} \cdot \underbrace{\Psi_t(y_{t-1} = v, y_t = u, x_t)}_{q(u|v) \cdot e(x_t|u)} \cdot \underbrace{p(x_{t+1}^T \mid y_t = u)}_{\beta_t(u)}$$

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Therefore equation (1) for $q(u|v)$ is:

$$\overline{q(u|v)} = \frac{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_t \alpha_{t-1}(v) \cdot q(u|v) \cdot e(x_t|u) \cdot \beta_t(u)}{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_t \alpha_t(u) \cdot \beta_t(u)}$$

For the parameter $e(b \mid u)$, we can express $P_\theta(x)$ as:

$$\sum_{\substack{t \\ x_t = b}} \sum_u \underbrace{p(x_1^t, y_t = u)}_{\alpha_t(u)} \cdot \underbrace{p(x_t^T \mid y_t = u)}_{\beta_t(u)}$$

For the parameter $e(b | u)$, we can express $P_\theta(x)$ as:

$$\sum_{\substack{t \\ x_t = b}} \sum_u \underbrace{p(x_1^t, y_t = u)}_{\alpha_t(u)} \cdot \underbrace{p(x_t^T | y_t = u)}_{\beta_t(u)}$$

Therefore equation (1) for $e(b|u)$ is:

$$\overline{e(b|u)} = \frac{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_{t; x_t=b} \alpha_t(u) \cdot \beta_t(u)}{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_t \alpha_t(u) \cdot \beta_t(u)} \quad (2)$$

Time complexity: $O(|x| |Q|)$

➤ **Definition:** Score for $x_1 \dots x_t$ starting at $y_t = s \in \mathcal{Y}$

$$\beta_t(s) \stackrel{\text{def}}{=} \sum_{y_{t+1}^T; y_t=s} \prod_{i=t+1}^T \Psi_i(y_{i-1}, y_i, x_i)$$

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$$\beta_t(s) \stackrel{\text{def}}{=} \sum_{y_{t+1}^T; y_t=s} \prod_{i=t+1}^T \Psi_i(y_{i-1}, y_i, x_i)$$

➤ **Initialization:** $\forall s \in \mathcal{Y}$

$$\beta_T(s) = 1$$

➤ **Recursion:** $\forall i = T - 1 \dots 1$; and $\forall s \in \mathcal{Y}$

$$\beta_t(s) = \sum_{s' \in \mathcal{Y}} \Psi_{t+1}(s, s', x_{t+1}) \cdot \beta_{t+1}(s')$$

➤ **Final result:** The optimal score for x is:

$$\sum_s \Psi_1(\text{null}, s, x_1) \cdot \beta_1(s)$$

Given a PCFG $(N, \Sigma, S, \mathcal{P}, \theta)$ in Chomsky Normal Form, where θ is a vector of parameters that contains two types of parameters for the two types of rules:

$$p(A \rightarrow BC) : A, B, C \in N \quad \text{and} \quad p(A \rightarrow b) : A \in N \quad b \in \Sigma$$



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$$p(A \rightarrow BC) : A, B, C \in N \text{ and } p(A \rightarrow b) : A \in N, b \in \Sigma$$

Then, Eq (1) becomes (see [Bened05]) :

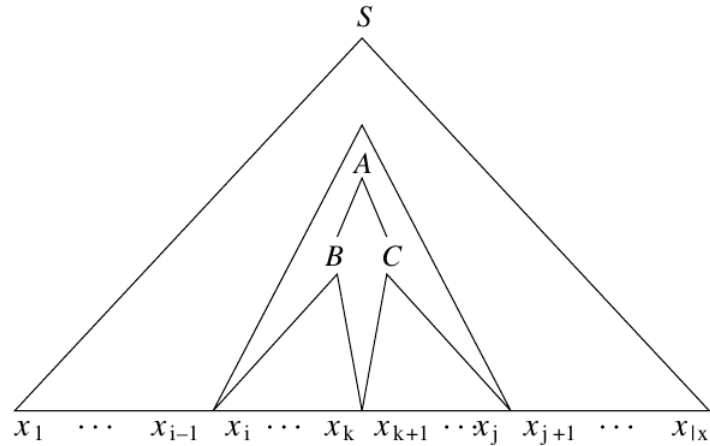
$$\bar{p}(A \rightarrow BC) = \frac{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_{\forall d_x} N(A \rightarrow BC, d_x) P_\theta(x, d_x)}{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_{\forall d_x} N(A, d_x) P_\theta(x, d_x)},$$

$$\bar{p}(A \rightarrow b) = \frac{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_{\forall d_x} N(A \rightarrow b, d_x) P_\theta(x, d_x)}{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_{\forall d_x} N(A, d_x) P_\theta(x, d_x)},$$

Note that the number of derivation trees can be exponential.

PCFGs: INSIDE-OUTSIDE ALGORITHM

Let $\Delta_{x,i,j,k,A \rightarrow BC}$ be the subset of derivations of x in which the rule $A \rightarrow BC$ appears delimited by positions i, j, k



For the parameters $p(A \rightarrow BC)$, we can express $P_\theta(x)$ as:

$$\sum_{\forall d_x} N(A \rightarrow BC, d_x) P_\theta(x, d_x)$$

$$= \sum_{A \rightarrow BC} \sum_{l=2}^T \sum_{i=0}^{T-l} \sum_{k=1}^{l-1} \underbrace{p(S \xRightarrow{*} x_1^i A x_{i+l+1}^T)}_{f(A,i,i+l)} \cdot \underbrace{p(B \xRightarrow{*} x_{i+1}^{i+k})}_{e(B,i,i+k)} \cdot \underbrace{p(C \xRightarrow{*} x_{i+k+1}^{i+l})}_{e(C,i+k,i+l)} \cdot p(A \rightarrow BC)$$

Therefore equation (1) for $p(A \rightarrow BC)$ is (see [Bened05] for alternative interpretation):

$$\bar{p}(A \rightarrow BC) = \frac{\sum_{x \in \mathcal{D}} \frac{p(A \rightarrow BC)}{P_{\theta}(x)} \sum_{l=2}^T \sum_{i=0}^{T-l} \sum_{k=1}^{l-1} f(A, i, i+l) \cdot e(B, i, i+k) \cdot e(C, i+k, i+l)}{\sum_{x \in \mathcal{D}} \frac{1}{P_{\theta}(x)} \sum_{l=1}^T \sum_{i=0}^{T-l} f(A, i, i+l) \cdot e(A, i, i+l)} \quad (3)$$

For the parameter $p(A \rightarrow b)$, we can express $P_\theta(x)$ as:



$$\sum_{A \rightarrow b} \sum_{i=0}^{T-1} \underbrace{p(S \xRightarrow{*} x_1^i A x_{i+2}^T)}_{f(A, i, i+1)} \cdot p(A \rightarrow b) \cdot \delta(b, x_{i+1})$$

Therefore equation (1) for $p(A \rightarrow b)$ is:

$$\bar{p}(A \rightarrow b) = \frac{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_{i=0}^{T-1} f(A, i, i+1) \cdot p(A \rightarrow b) \cdot \delta(b, x_{i+1})}{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x)} \sum_{l=1}^T \sum_{i=0}^{T-l} f(A, i, i+l) \cdot e(A, i, i+l)} \quad (4)$$

Time complexity: $O(|x|^3 |\mathcal{P}|)$ [Baker, 79; Lari and Young, 90; Bened05]

➤ **Definition:** Given $x = x_1 \dots x_T \in \Sigma^*$ and $A \in N$

$$f(A, i, j) = P_{G_p}(S \xRightarrow{*} x_1 \dots x_i A x_{j+1} \dots x_T)$$

➤ **Definition:** Given $x = x_1 \dots x_T \in \Sigma^*$ and $A \in N$

$$f(A, i, j) = P_{G_p}(S \xRightarrow{*} x_1 \dots x_i A x_{j+1} \dots x_T)$$

➤ **Initialization:** $\forall A \in N$

$$f(A, 0, T) = \delta(A, S)$$

➤ **Recursion:** $\forall A \in N; \quad \forall l : 1 \dots T; \quad \forall i : 0 \dots T - l;$

$$\begin{aligned} f(A, i, i + l) &= \sum_{B, C \in N} p(B \rightarrow CA) \sum_{k=1}^i f(B, i - k, i + l) \cdot e(C, i - k, i) \\ &+ \sum_{B, C \in N} p(B \rightarrow AC) \sum_{k=1}^{T-i-l} f(B, i, i + l + k) \cdot e(C, i + l, i + l + k) \end{aligned}$$

➤ **Final result:** The sentence probability is:

$$P_{G_p}(x) = \sum_{A \in N} f(A, i, i + 1) \cdot p(A \rightarrow x_{i+1}) \quad 0 \leq i \leq T - 1$$

Optimization function: Given a statistical model θ , and a training sample $\mathcal{D}$

$$\log P_{\theta}(\mathcal{D}, \hat{t}_{\mathcal{D}}) = \log \prod_{x \in \mathcal{D}} P_{\theta}(x, \hat{t}_x) = \sum_{x \in \mathcal{D}} \log P_{\theta}(x, \hat{t}_x)$$

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For the parameters $p(A \rightarrow \alpha)$, we can express $P_{\theta}(x, \hat{t}_x)$ as:

$$P_{\theta}(x, \hat{t}_x) = \prod_{(A \rightarrow \alpha)} p(A \rightarrow \alpha)^{N(A \rightarrow \alpha, \hat{t}_x)}$$

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$$P_{\theta}(x, \hat{t}_x) = \prod_{(A \rightarrow \alpha)} p(A \rightarrow \alpha)^{N(A \rightarrow \alpha, \hat{t}_x)}$$

Therefore equation (1) for $p(A \rightarrow \alpha)$ is:

$$\bar{p}(A \rightarrow \alpha) = \frac{\sum_{x \in \mathcal{D}} N(A \rightarrow \alpha, \hat{t}_x)}{\sum_{x \in \mathcal{D}} N(A, \hat{t}_x)}.$$

Time complexity: $O(|x| |\mathcal{P}|)$

[Ney, 91; Bened05]

Palindrome Language

$\{ ww^R \mid w \in \{a, b\}^+ \}$ w^R means the reverse string of w

➤ Original model

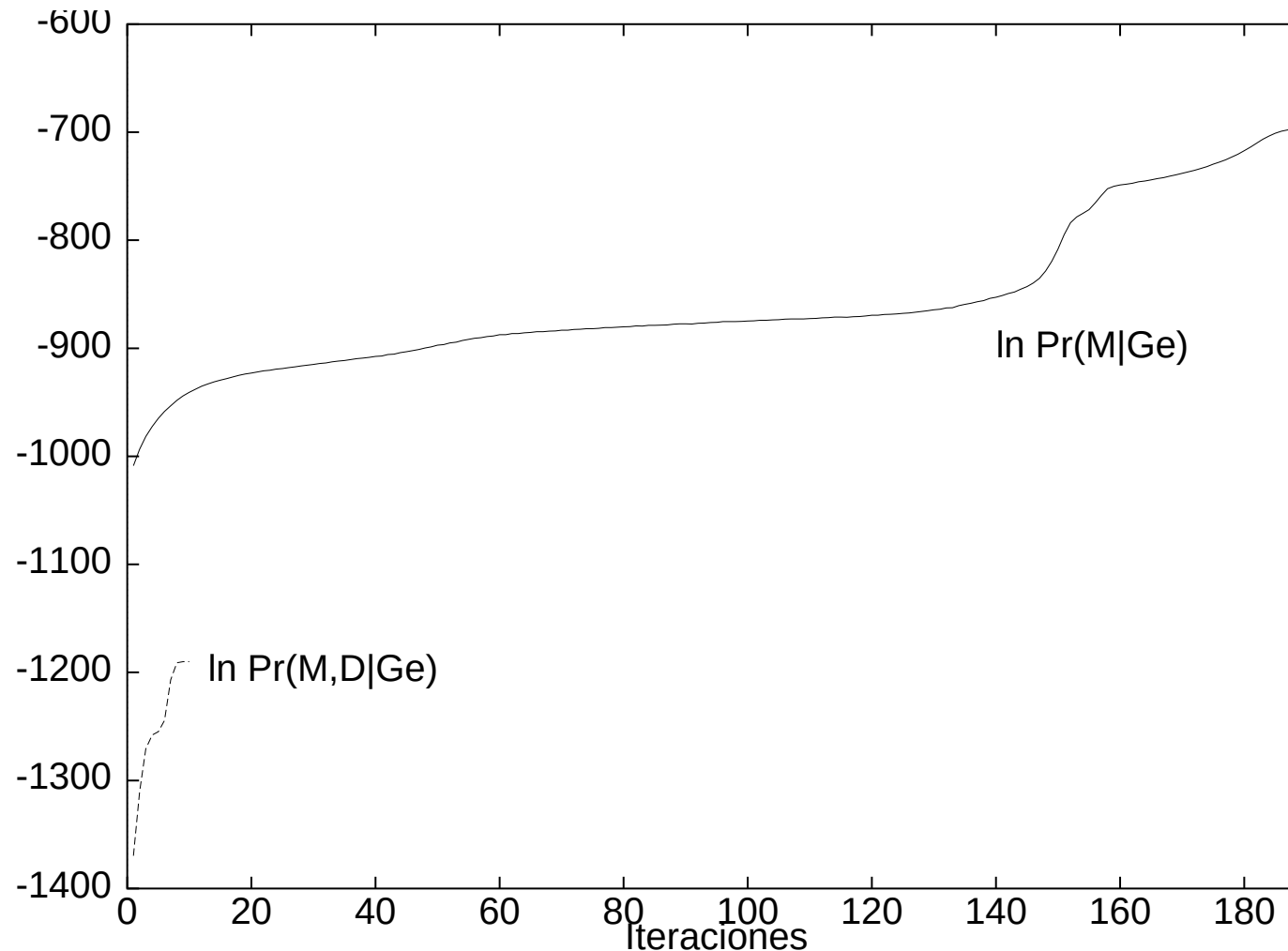
$S \rightarrow AC$ 0,4	$S \rightarrow BB$ 0,1	$C \rightarrow SA$ 1,0	$A \rightarrow a$ 1,0
$S \rightarrow BD$ 0,4	$S \rightarrow AA$ 0,1	$D \rightarrow SB$ 1,0	$B \rightarrow b$ 1,0

➤ Training sample: 1000 sentences

➤ Initial model

- 5 non-terminal symbols and 2 terminal symbols $\Rightarrow$ 130 rules
- Rule probabilities were randomly generated

Optimized functions with IO and VS algorithms



Accumulated probability mass

iteration	IO	
	apm	5bd
0	0.002	50.0 %
340	0.724	97.4 %

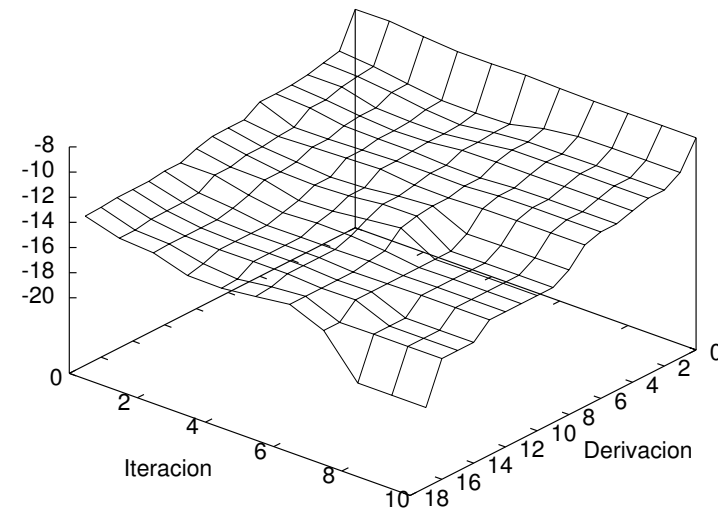
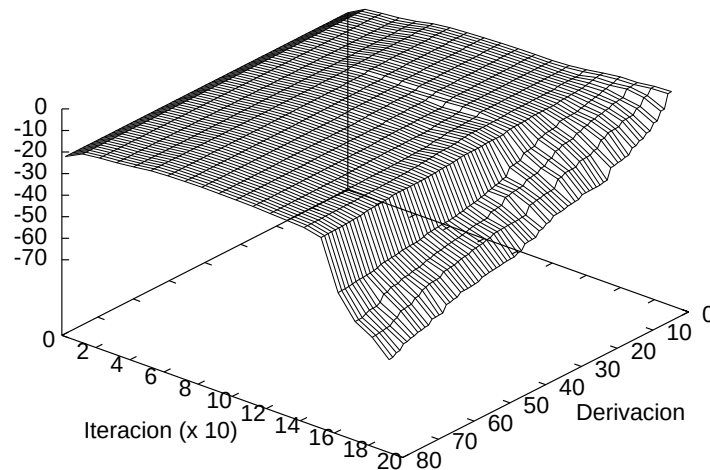
iteration	VS	
	apm	5bd
0	0.002	50.0 %
10	0.024	95.7 %

Accumulated probability mass

iteration	IO	
	apm	5bd
0	0.002	50.0 %
340	0.724	97.4 %

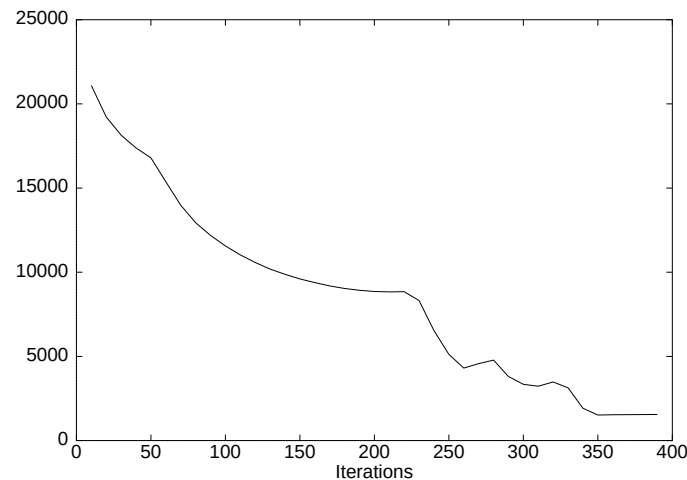
iteration	VS	
	apm	5bd
0	0.002	50.0 %
10	0.024	95.7 %

Evolution of the logarithm of the probability

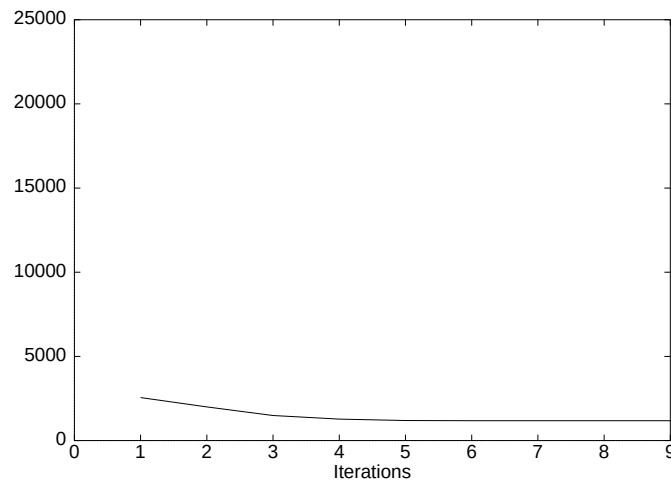


Upper bound estimate

IO



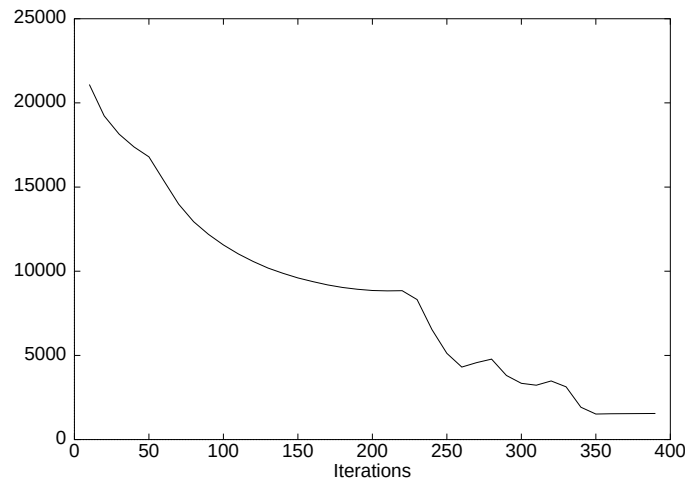
VS



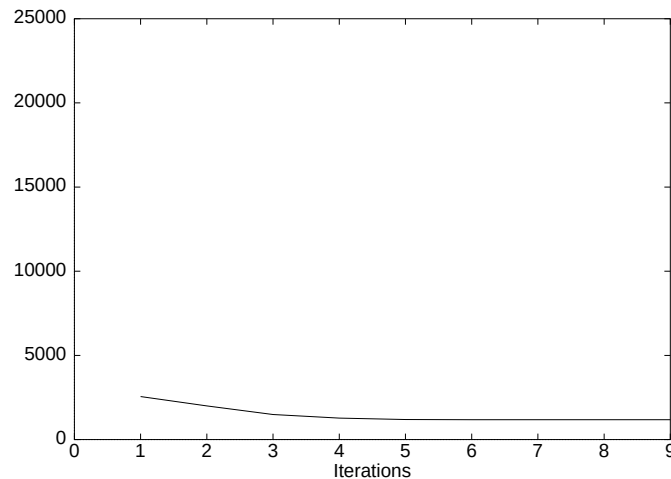
$$\log P_{\theta}(\mathcal{D}) - \log P_{\theta}(\mathcal{D}, \hat{t}_{\mathcal{D}}) \leq 2 \sum_{x \in \mathcal{D}} |x| \log 2 |N|$$

Upper bound estimate

IO



VS



$$\log P_{\theta}(\mathcal{D}) - \log P_{\theta}(\mathcal{D}, \hat{t}_{\mathcal{D}}) \leq 2 \sum_{x \in \mathcal{D}} |x| \log 2 |N|$$

Evaluation of models

Algorithm	kld	Palindromes (%)	Non palindromes (%)
VS	6.00	1.9	98.1
IO	1.88	76.0	24.0

Optimization function: Given G_θ , and a training sample $\mathcal{D}$

$$\lg P_\theta(\mathcal{D}, \Delta_{\mathcal{D}}) = \lg \prod_{x \in \mathcal{D}} P_\theta(x, \Delta_x) = \sum_{x \in \mathcal{D}} \lg P_\theta(x, \Delta_x)$$

Optimization function: Given G_θ , and a training sample $\mathcal{D}$

$$\lg P_\theta(\mathcal{D}, \Delta_{\mathcal{D}}) = \lg \prod_{x \in \mathcal{D}} P_\theta(x, \Delta_x) = \sum_{x \in \mathcal{D}} \lg P_\theta(x, \Delta_x)$$

For the parameters $p(A \rightarrow \alpha)$, we can express $P_\theta(x, \Delta_x)$ as:

$$P_\theta(x, \Delta_x) = \sum_{t_x \in \Delta_x} P_\theta(x, t_x) = \sum_{t_x \in \Delta_x} \prod_{(A \rightarrow \alpha)} p(A \rightarrow \alpha)^{N(A \rightarrow \alpha, t_x)}$$

Optimization function: Given G_θ , and a training sample $\mathcal{D}$

$$\lg P_\theta(\mathcal{D}, \Delta_{\mathcal{D}}) = \lg \prod_{x \in \mathcal{D}} P_\theta(x, \Delta_x) = \sum_{x \in \mathcal{D}} \lg P_\theta(x, \Delta_x)$$

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Therefore equation (1) for $p(A \rightarrow \alpha)$ is:

$$\bar{p}(A \rightarrow \alpha) = \frac{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x, \Delta_x)} \sum_{t_x \in \Delta_x} N(A \rightarrow \alpha, t_x) \cdot P_\theta(x, t_x)}{\sum_{x \in \mathcal{D}} \frac{1}{P_\theta(x, \Delta_x)} \sum_{t_x \in \Delta_x} N(A, t_x) \cdot P_\theta(x, t_x)} \quad (5)$$

Time complexity: $O(|x|^3 |\mathcal{P}|)$

[BenedShez 05]

➤ k VS algorithm:

1. Obtaining the k -best derivations
2. Applying the transformation

➤ Calculation of k -best derivations [Jimz et al. 2000]

⇒ Time complexity

➤ Worst case: $O(|x|^3|\mathcal{P}| + k|x|\log k + |x|^3|\mathcal{P}|\log k)$

➤ Best case: $O(|x|^3|\mathcal{P}| + k + \frac{|x||\mathcal{P}|}{|N|})$

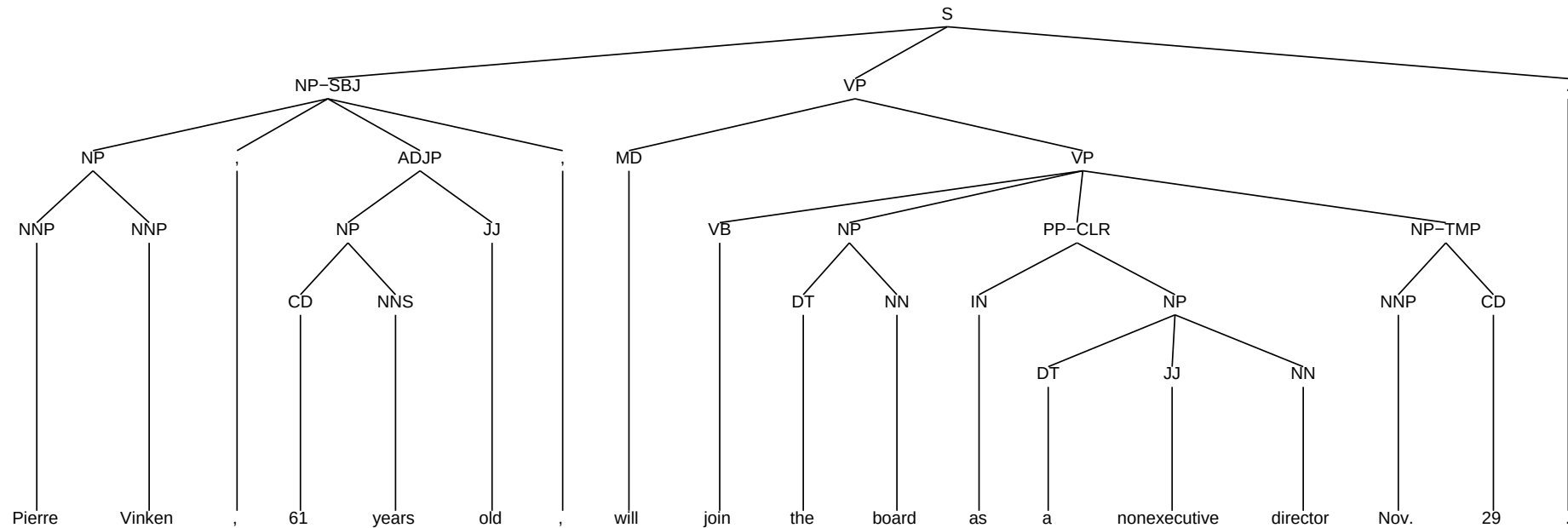
⇒ Space complexity

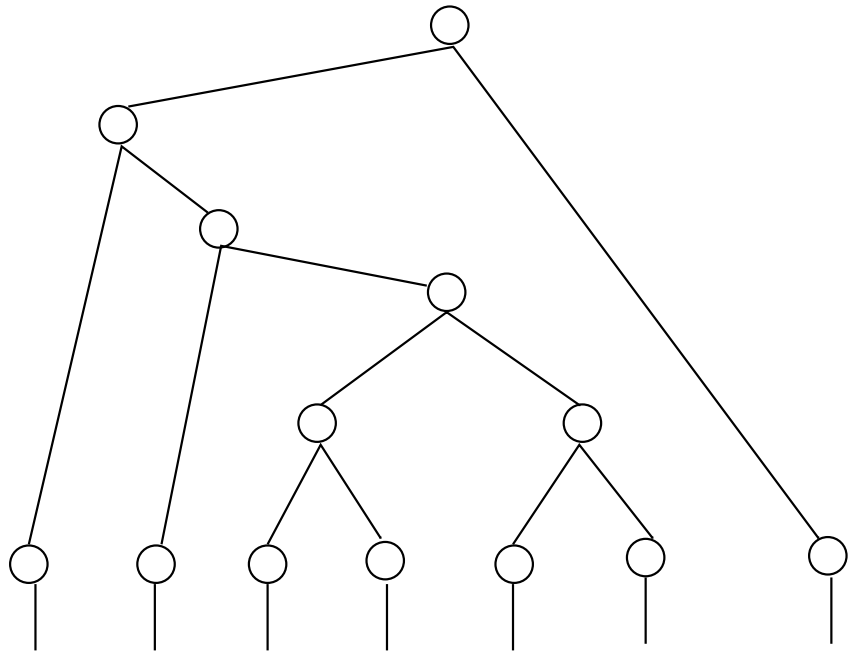
➤ $O(k(|x|^2|N| + |x||N|))$

Estimation algorithm	kld	Palindromes		Non palindromes	
		%	apm	%	apm
VS	6.00	1.9	0.023	98.1	0.851
kVS ($k = 2$)	5.76	2.3	0.028	97.7	0.835
kVS ($k = 4$)	5.68	3.7	0.040	96.3	0.822
kVS ($k = 8$)	5.56	7.9	0.074	92.1	0.800
kVS ($k = 16$)	5.45	7.8	0.076	92.2	0.793
kVS ($k = 32$)	5.07	13.2	0.111	86.8	0.696
kVS ($k = 64$)	5.03	12.7	0.112	87.3	0.691
kVS ($k = 128$)	4.98	11.2	0.095	88.8	0.685
kVS ($k = 256$)	4.95	11.9	0.119	88.1	0.637
kVS ($k = 512$)	3.59	17.6	0.156	82.4	0.372
kVS ($k = 1024$)	3.59	17.9	0.156	82.1	0.373
kVS ($k = 2048$)	3.59	17.9	0.156	82.1	0.373
IO	1.88	76.0	0.710	24.0	0.044

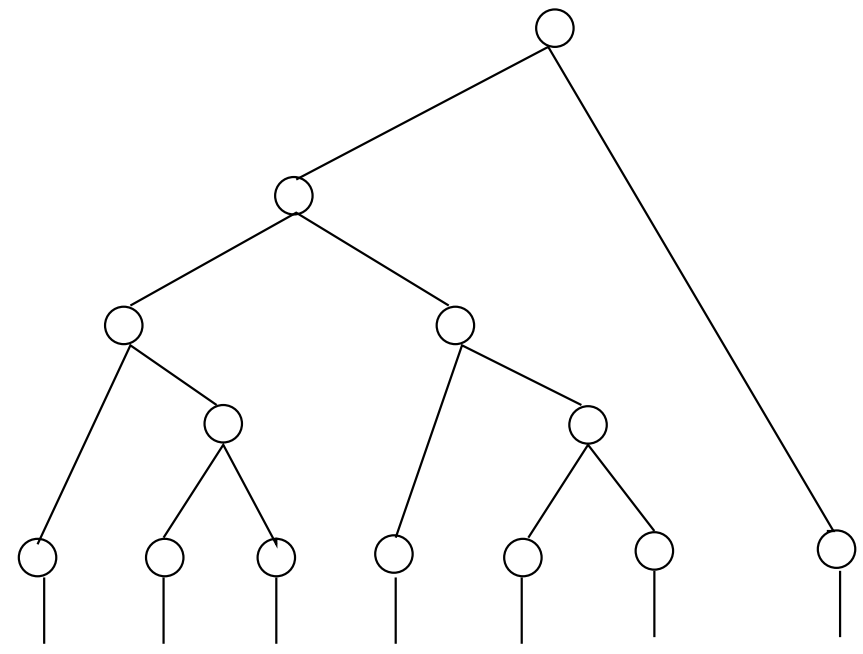
Example: *“Pierre Vinken, 61 years old, will join the board as a nonexecutive director Nov. 29.”*

((S (NP-SBJ (NP (NNP Pierre) (NNP Vinken)) (,,) (ADJP (NP (CD 61) (NNS years)) (JJ old)) (,,)) (VP (MD will) (VP (VB join) (NP (DT the) (NN board)) (PP-CLR (IN as) (NP (DT a) (JJ nonexecutive) (NN director)))) (NP-TMP (NNP Nov.) (CD 29)))) (..)))

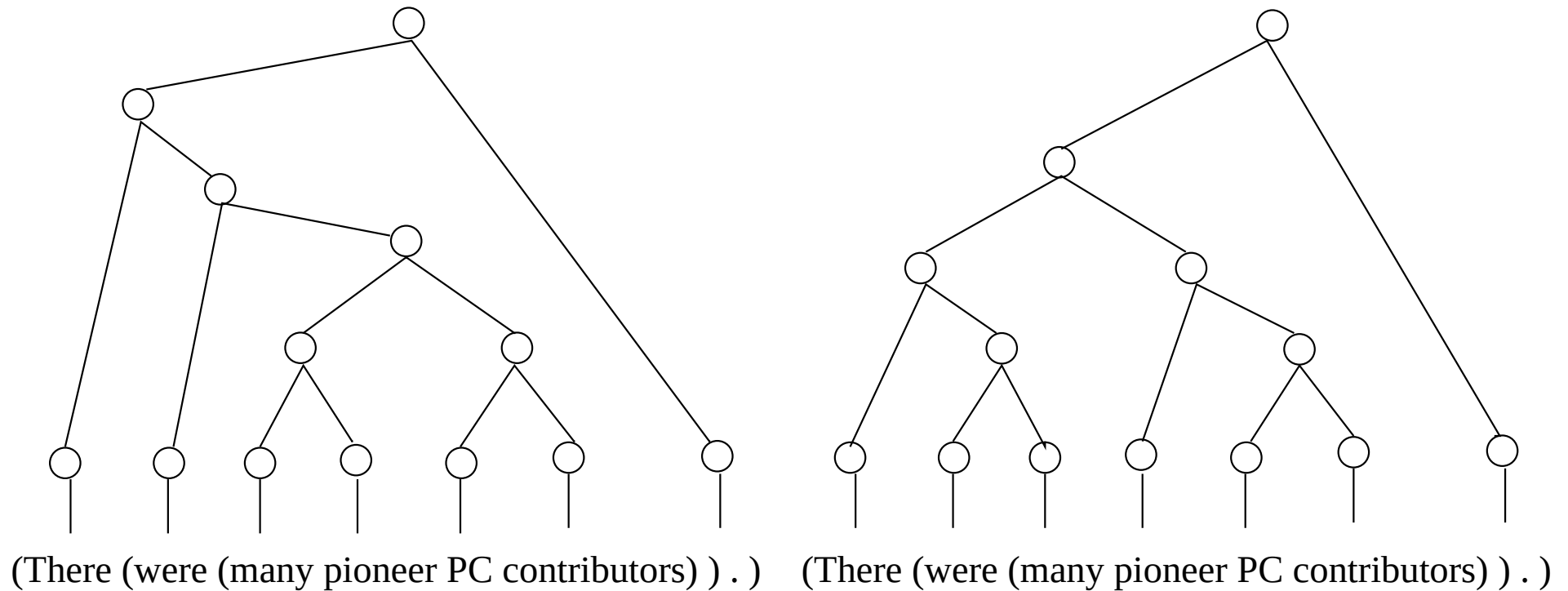




(There (were (many pioneer PC contributors)) .)



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From a set of bracketed corpus, the following function is defined:

[Pereira and Schabes, 1992]

$$\bar{c}(i, j) = \begin{cases} 1 & \text{if } (i, j) \text{ is consistent with the bracketed sample,} \\ 0 & \text{otherwise} \end{cases}$$

(Bracketed) *Viterbi* algorithm

[Pereira and Schabes, 1992]

$\forall A \in N$, we compute:

$$\hat{e}^{(b)}(A, i, i+1) = p(A \rightarrow b) \cdot \delta(b, x_{i+1}) \quad 0 \leq i < T-1$$

$$\begin{aligned} \hat{e}^{(b)}(A, i, i+l) = & \bar{c}(i, i+l) \max_{B, C \in N} \cdot p(A \rightarrow BC) \\ & \max_{k=1, \dots, l-1} \hat{e}^{(b)}(B, i, i+k) \hat{e}^{(b)}(C, i+k, i+l) \end{aligned}$$

$$2 \leq l \leq T; 0 \leq i \leq T-l$$

$$\hat{P}_\theta(x) = \hat{e}^{(b)}(S < 0, n >).$$

Estimation algorithm	kld	Palindromes		Non palindromes	
		%	apm	%	apm
VSb	4.49	17.1	0.159	82.9	0.600
kVSb ($k = 2$)	4.96	12.8	0.119	87.2	0.633
kVSb ($k = 4$)	4.49	16.8	0.159	83.2	0.600
kVSb ($k = 8$)	4.47	18.2	0.158	81.8	0.591
kVSb ($k = 16$)	4.45	17.6	0.162	82.4	0.584
kVSb ($k = 32$)	4.43	16.9	0.170	83.1	0.578
kVSb ($k = 64$)	4.41	17.5	0.174	82.5	0.572
kVSb ($k = 128$)	3.93	22.5	0.200	77.5	0.564
kVSb ($k = 256$)	3.80	31.6	0.303	68.4	0.442
kVSb ($k = 512$)	3.23	27.1	0.264	72.9	0.299
kVSb ($k = 1024$)	3.59	17.6	0.157	82.4	0.370
kVSb ($k = 2048$)	1.88	74.7	0.715	25.3	0.049
IOb	0.00	100.0	0.746	0.0	0.000

➤ Characteristics of WSJ corpus:

Data set	Directories	No. of senten.	No. of words
Training (full)	00-20	42,075	1,004,073
Training (≤ 50)	00-20	41,315 (98,2 %)	959,390 (95,6 %)
Tuning	21-22	3,371	80,156
Test	23-24	3,762	89,537

➤ Vocabulary (Training), only the 10,000 most frequent words

➤ Category-based PCFGs (length restriction ≤ 50):

$ N $	(size)	VS		kVS		VSb		$kVSb$		IOb	
		tsp	size	($k = 128$) tsp	size	tsp	size	($k = 128$) tsp	size	tsp	size
14	(3,374)	20.13	195	18.56	195	21.10	256	19.53	256	13.34	471
20	(8,900)	20.42	255	17.46	280	19.38	406	18.01	311	12.33	888
25	(16,750)	17.58	407	16.76	349	16.93	499	16.89	398	11.86	1042
30	(28,350)	17.67	367	16.34	400	17.90	609	16.78	424	10.86	1454
35	(44,450)	16.92	600	16.02	454	16.64	789	16.24	619	10.24	1741

The goal of supervised (or predictive) learning is to learn a mapping from inputs $\boldsymbol{x}$ to outputs $\boldsymbol{y}$, given a fully annotated training set $\mathcal{D} = \{(\boldsymbol{x}^{(n)}, \boldsymbol{y}^{(n)})\}_{n=1}^N$.

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➤ Objective function: Maximum likelihood (generative) estimation [Bahl et al. 1983]

$$\log \prod_{n=1}^N P_{\theta}(\mathbf{x}^{(n)}, \mathbf{y}^{(n)}) = \sum_{n=1}^N \log P_{\theta}(\mathbf{x}^{(n)}, \mathbf{y}^{(n)})$$

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$$P_{\theta}(x^{(n)}, y^{(n)}) = \prod_{\theta_{ij}} \theta_{ij}^{N(\theta_{ij}, (x^{(n)}, y^{(n)}))}$$

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➤ **Optimization methods: Growth transformations**

$$\bar{\theta}_{ij} = \frac{\sum_{n=1}^N \frac{\theta_{ij}}{P_{\theta}(\mathbf{x}^{(n)}, \mathbf{y}^{(n)})} \frac{\partial P_{\theta}(\mathbf{x}^{(n)}, \mathbf{y}^{(n)})}{\partial \theta_{ij}}}{\sum_{n=1}^N \frac{1}{P_{\theta}(\mathbf{x}^{(n)}, \mathbf{y}^{(n)})} \sum_k \theta_{ik} \frac{\partial P_{\theta}(\mathbf{x}^{(n)}, \mathbf{y}^{(n)})}{\partial \theta_{ik}}} = \frac{\sum_{n=1}^N \theta_{ij} N(\theta_{ij}, (\mathbf{x}^{(n)}, \mathbf{y}^{(n)}))}{\sum_{n=1}^N \sum_k \theta_{ik} N(\theta_{ik}, (\mathbf{x}^{(n)}, \mathbf{y}^{(n)}))} \quad (6)$$

Discriminative Training: Objective functions

- Maximum conditional likelihood (discriminative) estimation [Jebara & Pentland, 1998]

$$F_{\theta}(\mathcal{D}) = \prod_{n=1}^N P_{\theta}(y^{(n)}|x^{(n)}) = \prod_{n=1}^N \frac{P_{\theta}(x^{(n)}, y^{(n)})}{P_{\theta}(x^{(n)})} = \prod_{n=1}^N \frac{P_{\theta}(x^{(n)}|y^{(n)}) P_{\theta}(y^{(n)})}{\sum_{y \in \theta(x^{(n)})} P_{\theta}(x^{(n)}|y) P_{\theta}(y)}$$

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- Maximum mutual information (discriminative) estimation [Bahl, et al. 1986]
- Minimum classification error [Juang, et al. 1997]
- Large Margin Estimation [Jiang, 2004]

Discriminative Training: Objective functions

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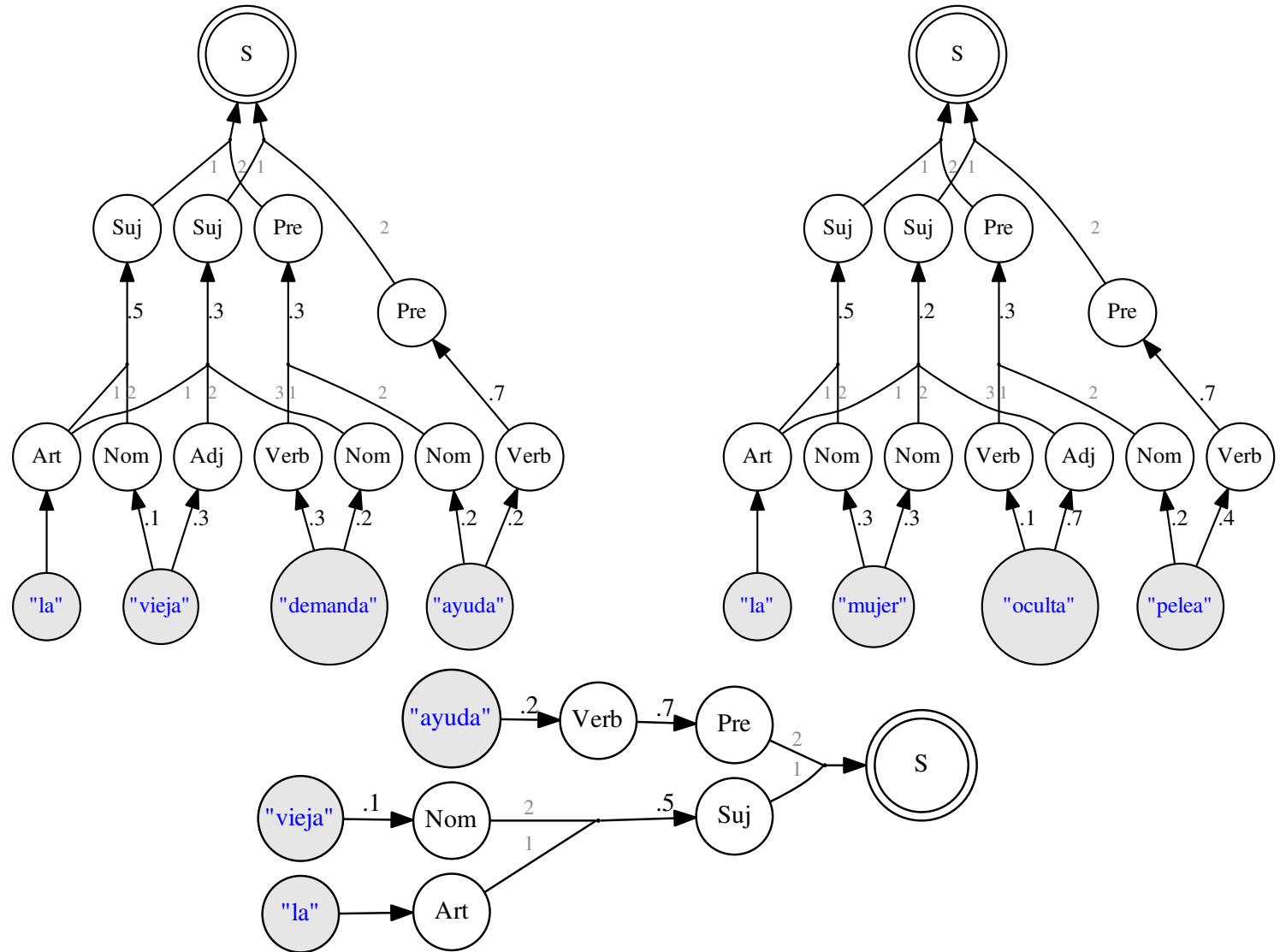
- Maximum mutual information (discriminative) estimation [Bahl, et al. 1986]
- Minimum classification error [Juang, et al. 1997]
- Large Margin Estimation [Jiang, 2004]

Discriminative Training: Optimization methods

- Gradient Descent
- Extended Baum method (Growth transformations) [Gopalakrishnan et al. 1991]

Grammar: (G, θ)

- 1.0 S $\rightarrow$ Suj Pre
- 0.5 Suj $\rightarrow$ Art Nom
- 0.3 Suj $\rightarrow$ Art Adj Nom
- 0.2 Suj $\rightarrow$ Art Nom Adj
- 0.3 Pre $\rightarrow$ Verb Nom
- 0.7 Pre $\rightarrow$ Verb
- 1.0 Art $\rightarrow$ "la"
- 0.1 Nom $\rightarrow$ "vieja"
- 0.2 Nom $\rightarrow$ "ayuda"
- 0.3 Nom $\rightarrow$ "mujer"
- 0.2 Nom $\rightarrow$ "pelea"
- 0.2 Nom $\rightarrow$ "demanda"
- 0.3 Verb $\rightarrow$ "demanda"
- 0.2 Verb $\rightarrow$ "ayuda"
- 0.1 Verb $\rightarrow$ "oculta"
- 0.4 Verb $\rightarrow$ "pelea"
- 0.3 Adj $\rightarrow$ "vieja"
- 0.7 Adj $\rightarrow$ "oculta"



Sample: $\mathcal{D} = \{\text{la vieja demanda ayuda, la mujer oculta pelea, la vieja ayuda}\}$

$$P_{\theta}(\text{la vieja demanda ayuda}) = ,00090 + ,00252 = ,00342$$

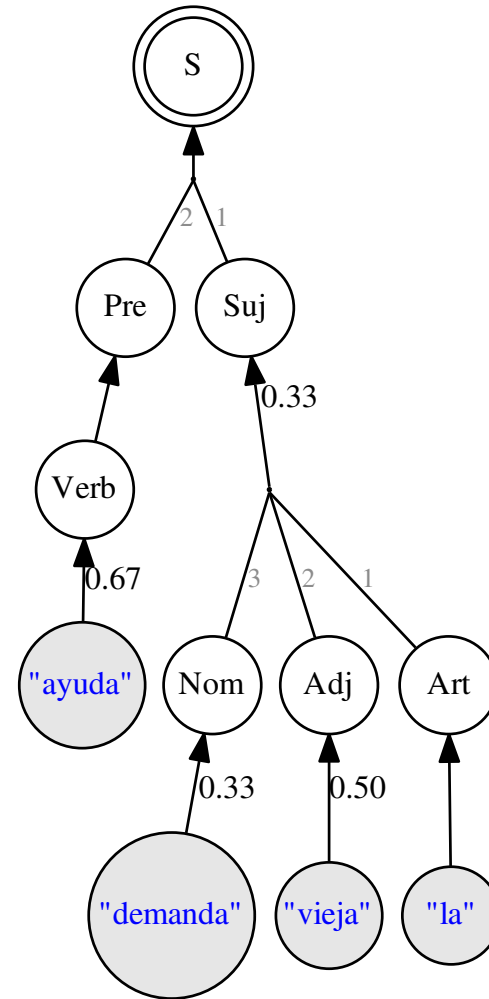
$$P_{\theta}(\text{la mujer oculta pelea}) = ,00090 + ,01176 = ,01266$$

$$P_{\theta}(\text{la vieja ayuda}) = ,00700$$

Inside-Outside:

$$\begin{aligned} & \bar{p}(\text{Suj} \rightarrow \text{Art Nom Adj}) \\ \text{☰} \quad &= \frac{\sum_{x \in \mathcal{D}} \frac{1}{P_{\theta}(x)} \sum_{t_x} N(\text{Suj} \rightarrow \text{Art Nom Adj}, t_x) P_{\theta}(x, t_x)}{\sum_{x \in \mathcal{D}} \frac{1}{P_{\theta}(x)} \sum_{t_x} N(\text{Suj}, t_x) P_{\theta}(x, t_x)} \\ &= \frac{\frac{1}{,01266} ,01176}{\frac{1}{,00342} (,00090 + ,00252) + \frac{1}{,01266} (,00090 + ,01176) + \frac{1}{,00700} (,00700)} \\ &= \frac{,9289}{3} = ,3096 \end{aligned}$$

Inside-Outside:	θ_1	θ_2	θ_3
1.0 S $\rightarrow$ Suj Pre	1.0000	1.0000	1.0000
0.5 Suj $\rightarrow$ Art Nom	0.4447	0.3370	0.3333
0.3 Suj $\rightarrow$ Art Adj Nom	0.2456	0.3299	0.3333
0.2 Suj $\rightarrow$ Art Nom Adj	0.3096	0.3331	0.3334
0.3 Pre $\rightarrow$ Verb Nom	0.1114	0.0036	
0.7 Pre $\rightarrow$ Verb	0.8886	0.9964	1.0000
1.0 Art $\rightarrow$ "la"	1.0000	1.0000	1.0000
0.1 Nom $\rightarrow$ "vieja"	0.3788	0.3356	0.3333
0.2 Nom $\rightarrow$ "ayuda"	0.0789	0.0034	
0.3 Nom $\rightarrow$ "mujer"	0.2999	0.3322	0.3333
0.2 Nom $\rightarrow$ "pelea"	0.0213	0.0002	
0.2 Nom $\rightarrow$ "demanda"	0.2201	0.3286	0.3334
0.3 Verb $\rightarrow$ "demanda"	0.0877	0.0035	
0.2 Verb $\rightarrow$ "ayuda"	0.5790	0.6632	0.6667
0.1 Verb $\rightarrow$ "oculta"	0.0237	0.0002	
0.4 Verb $\rightarrow$ "pelea"	0.3096	0.3331	0.3333
0.3 Adj $\rightarrow$ "vieja"	0.4423	0.4976	0.5000
0.7 Adj $\rightarrow$ "oculta"	0.5577	0.5024	0.5000

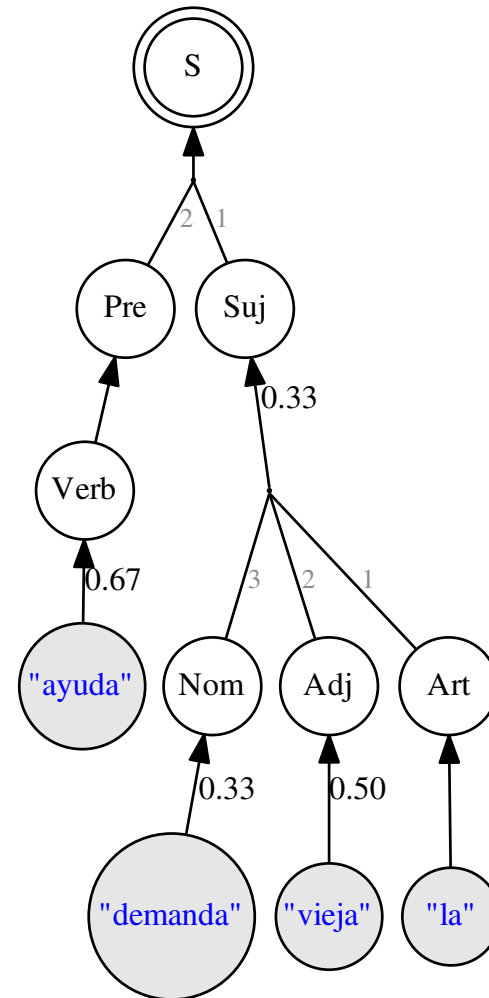


$$L_{\Theta}(\mathcal{D}) = 3,0 \cdot 10^{-7} \quad 8,7 \cdot 10^{-6} \quad 4,9 \cdot 10^{-5} \quad 5,1 \cdot 10^{-5}$$

Viterbi:

θ_1

1.0 S	→ Suj Pre	1.0000
0.5 Suj	→ Art Nom	0.3333
0.3 Suj	→ Art Adj Nom	0.3333
0.2 Suj	→ Art Nom Adj	0.3334
0.3 Pre	→ Verb Nom	
0.7 Pre	→ Verb	1.0000
1.0 Art	→ "la"	1.0000
0.1 Nom	→ "vieja"	0.3333
0.2 Nom	→ "ayuda"	
0.3 Nom	→ "mujer"	0.3333
0.2 Nom	→ "pelea"	
0.2 Nom	→ "demanda"	0.3334
0.3 Verb	→ "demanda"	
0.2 Verb	→ "ayuda"	0.6667
0.1 Verb	→ "oculta"	
0.4 Verb	→ "pelea"	0.3333
0.3 Adj	→ "vieja"	0.5000
0.7 Adj	→ "oculta"	0.5000



$$L_{\Theta}(\mathcal{D}) = 3,0 \cdot 10^{-7}$$

$$8,7 \cdot 10^{-6} \quad 4,9 \cdot 10^{-5} \quad 5,1 \cdot 10^{-5}$$

- R.Schlter, W.Macherey, B.Mller, H.Ney (2001) **Comparison of discriminative training criteria and optimization methods for speech recognition**. Speech Communication (34):287-310
- H.Jiang (2010) **Discriminative training of HMMs for automatic speech recognition: A survey**. Computer Speech and Language (24):589-608.

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➤ Discriminative models: **Conditional Random Fields (CRF)**

[Lafferty, McCallum, Pereira, 2001].

$$\begin{aligned} P_{\theta}(y|x) &= \frac{\exp\{ \phi(x, y; \theta) \}}{Z(x; \theta)} \\ &= \frac{\exp\{ \sum_{t=1}^T \sum_{k=1}^K \theta_k f_k(y_{t-1}, y_t, x_t) \}}{Z(x; \theta)} \end{aligned} \quad (7)$$

Where

$$Z(x; \theta) = \sum_{y' \in y^*} \exp \left\{ \sum_{t=1}^T \sum_{k=1}^K \theta_k f_k(y'_{t-1}, y'_t, x_t) \right\} \quad (8)$$

- Given $\mathcal{D} = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(N)}, y^{(N)})\}$, finding the optimal parameters of a CRF:

$$\hat{\theta} = \arg \max_{\theta} L_{\theta}(\mathcal{D})$$

- Objective function: Conditional log likelihood

$$L_{\theta}(\mathcal{D}) = \sum_{n=1}^N \log P_{\theta}(y^{(n)} | x^{(n)}) \quad (9)$$

- Given $\mathcal{D} = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(N)}, y^{(N)})\}$, finding the optimal parameters of a CRF:

$$\hat{\theta} = \arg \max_{\theta} L_{\theta}(\mathcal{D})$$

- Objective function: Conditional log likelihood

$$L_{\theta}(\mathcal{D}) = \sum_{n=1}^N \log P_{\theta}(y^{(n)} | x^{(n)}) \quad (10)$$

- Objective function: Regularized log likelihood

$$L_{\theta}(\mathcal{D}) = \sum_{n=1}^N \log P_{\theta}(y^{(n)} | x^{(n)}) - \frac{\lambda}{2} \sum_k \theta_k^2 \quad (11)$$

The second term is an additional regularization term, with $\lambda > 0$, to control the trade-off between fitting to data and model complexity

- Substituting the definitions of CRF model (7, 8) into the expression of $L_\theta(\mathcal{D})$ (10)

$$L_\theta(\mathcal{D}) = \sum_{n=1}^N \sum_{t=1}^T \sum_{k=1}^K \theta_k f_k(y_{t-1}^{(n)}, y_t^{(n)}, x_t^{(n)}) - \sum_{n=1}^N \log Z(x^{(n)}; \theta) \quad (12)$$

- Find

$$\hat{\theta} = \arg \max_{\theta} L_\theta(\mathcal{D}) - \frac{\lambda}{2} \sum_k \theta_k^2$$

- In general there is no analytical solution to this optimization, so we must use a numerical optimization, i.e. gradient-based optimization.
- It can be shown that $L_\theta(\mathcal{D})$ is a concave function. It means that every local optimum is also a global optimum. Adding regularization ensures that $L_\theta(\mathcal{D})$ is strictly concave, which implies that it has exactly one global optimum [Sutton & McCallum, 2010].

➤ Gradient-based optimization

1 Initialize: $\theta = 0$

2 Compute the gradient:
$$\delta_k = \frac{\partial L_{\theta}(\mathcal{D})}{\partial \theta_k} \quad \forall k = 1 \dots K$$

3 Calculate
$$\hat{\beta} = \arg \max_{\beta} L_{(\theta + \beta \delta)}(\mathcal{D})$$

4 Move θ in steps proportional to the gradient:

$$\theta = \theta + \hat{\beta} \delta$$

5 Repeat steps 2 to 4 until convergence

- The step 3 can be performed using some type of linear search.
- There are many standard packages which use more sophisticated algorithms, and which give considerably faster convergence.

$$\begin{aligned} \frac{\partial L(\theta)}{\partial \theta_k} &= \sum_{n=1}^N \sum_{t=1}^T f_k(y_{t-1}^{(n)}, y_t^{(n)}, x_t^{(n)}) \\ &\quad - \sum_{n=1}^N \sum_{t=1}^T \sum_{y \in \mathcal{Y}^T} f_k(y_{t-1}, y_t, x_t^{(n)}) P_{\theta}(y | x^{(n)}) \end{aligned} \quad (13)$$

- The first term is simply the number of times that the feature f_k is equal to 1 on the training data. This term is easy to compute by counting.
- The second term can be interpreted as the expected number of times the feature f_k is equal to 1. This term is more involved to compute.
- For the Regularized log likelihood (11), add the term: $-\lambda \theta_k$

$$\begin{aligned} \sum_{t=1}^T \sum_{y \in \mathcal{Y}^T} f_k(y_{t-1}, y_t, x_t^{(n)}) P_\theta(y | x^{(n)}) &= \\ \sum_{t=1}^T \sum_{s, s' \in \mathcal{Y}} f_k(y_{t-1} = s', y_t = s, x_t^{(n)}) \sum_{\substack{y \in \mathcal{Y}^T \\ y_{t-1} = s', y_t = s}} P_\theta(y | x^{(n)}) & \quad (14) \end{aligned}$$

From the definition of CRF (7):

$$P_\theta(y|x) = \frac{1}{Z(x; \theta)} \prod_{t=1}^T \Psi_t(y_{t-1}, y_t, x_t)$$

Where:

$$\Psi_t(y_{t-1}, y_t, x_t) = \exp \left\{ \sum_{k=1}^K \theta_k f_k(y_{t-1}, y_t, x_t) \right\}$$

➤ Forward variables

$$\alpha_t(s) = \sum_{y_1^t; y_t=s} \prod_{j=1}^t \Psi_j(y_{j-1}, y_j, x_j)$$

➤ Backward variables

$$\beta_t(s) = \sum_{y_{t+1}^T; y_t=s} \prod_{j=t+1}^T \Psi_j(y_{j-1}, y_j, x_j)$$

➤ The quantity Z

$$Z = \sum_s \alpha_T(s)$$

Finally, the marginal distribution needed for the gradient can be computed as:

$$\sum_{\substack{y \in \mathcal{Y}^T \\ y_{t-1} = s', y_t = s}} P_\theta(y | x^{(n)}) = \alpha_{t-1}(s') \Psi_t(y_{t-1} = s', y_t = s, x_t^{(n)}) \beta_t(s) Z^{-1}$$

➤ Given a training corpus $\mathcal{D} = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(N)}, y^{(N)})\}$

Goal: to learn a predictor $x \rightarrow y$ with small error on unseen inputs.

➤ In CRFs:

$$\begin{aligned} \arg \max_y P_\theta(y|x) &= \arg \max_y \frac{\exp\{ \sum_{t=1}^T \sum_{k=1}^K \theta_k f_k(y_{t-1}, y_t, x_t) \}}{Z(x; \theta)} \\ &= \arg \max_y \sum_{t=1}^T \theta f(y_{t-1}, y_t, x_t) \end{aligned}$$

➤ Can we learn θ more directly, focusing on errors?

[Collins, 2002]

- Given a training set $\mathcal{D} = \{(x^{(n)}, y^{(n)})\}_{n=1}^N$
- Set $\theta = 0$
- For each iteration $w : 1 \dots W$, and for each training sample $(x^{(n)}, y^{(n)}) : n : 1 \dots N$

- Compute

$$z^{(n)} = \arg \max_z \sum_{t=1}^T \theta f(z_{t-1}, z_t, x_t^{(n)})$$

- if $z^{(n)} \neq y^{(n)}$

$$\theta = \theta + \sum_{t=1}^T f(y_{t-1}^{(n)}, y_t^{(n)}, x_t^{(n)}) - \sum_{t=1}^T f(z_{t-1}^{(n)}, z_t^{(n)}, x_t^{(n)})$$

- Return θ

[Collins, 2002] approximation of voted perceptron [Freund & Schapire, 1999]

➤ Given a training set $\mathcal{D} = \{(x^{(n)}, y^{(n)})\}_{i=1}^N$

➤ Set $\theta = 0$; $\theta_a = 0$

➤ For each iteration $w : 1 \dots W$, and for each training sample $(x^{(n)}, y^{(n)}) : n : 1 \dots N$

➤ Compute

$$z^{(n)} = \arg \max_z \sum_{t=1}^T \theta f(z_{t-1}, z_t, x_t^{(n)})$$

➤ if $z^{(n)} \neq y^{(n)}$

$$\theta = \theta + \sum_{t=1}^T f(y_{t-1}^{(n)}, y_t^{(n)}, x_t^{(n)}) - \sum_{t=1}^T f(z_{t-1}^{(n)}, z_t^{(n)}, x_t^{(n)})$$

➤ $\theta_a = \theta_a + \theta$

➤ Return $\theta_a / N \cdot W$

- If the data is separable, Perceptron algorithm converges.
Also there are partial solutions for non-separable cases.
- It is a very simple framework that can work with many structured problems and that works very well:
 - all you need is (fast) 1-best inference (Viterbi);
 - typically reaches a good solution after only a few iterations;
 - can be applied to parsing, translation, etc.

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- J.R.Finkel, A.Kleeman and C.D.Manning: *Efficient, Feature-based, Conditional Random Field Parsing*. ACL/HLT, pp. 959–967, 2008.

Problem definition [Liang 09]:

- Given a set of unlabeled examples $\mathcal{D} = \{\mathbf{x}^{(n)}\}_{n=1}^N$, maximizing the log-likelihood:

$$l(\theta) = \sum_{n=1}^N \log P_{\theta}(\mathbf{x}^{(n)})$$

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- Training algorithm: **EM algorithm** [Dempster 77, Neal 98, Capp9]

$$\overline{\theta}_{ij} = \frac{\sum_{n=1}^N \frac{1}{P_{\theta}(\mathbf{x}^{(n)})} \sum_{\mathbf{t} \in \theta(\mathbf{x}^{(n)})} \mathcal{N}(\theta_{ij}, \mathbf{t}) P_{\theta}(\mathbf{x}^{(n)}, \mathbf{t})}{\sum_{n=1}^N \frac{1}{P_{\theta}(\mathbf{x}^{(n)})} \sum_{\mathbf{t} \in \theta(\mathbf{x}^{(n)})} \mathcal{N}(\theta_i, \mathbf{t}) P_{\theta}(\mathbf{x}^{(n)}, \mathbf{t})} \quad \forall \theta_{ij}$$

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Rewriting the numerator for all parameters:

$$\sum_{n=1}^N \sum_{\mathbf{t} \in \theta(\mathbf{x}^{(n)})} \phi(\mathbf{x}^{(n)}, \mathbf{t}) P_{\theta}(\mathbf{t} | \mathbf{x}^{(n)})$$

EM algorithm [Liang 09]:

Batch EM

```
 $\theta \leftarrow \text{initialization}$   
for each iteration  $w = 1, \dots, W$ :  
   $\mu \leftarrow 0$   
  for each example  $n = 1, \dots, N$ :  
     $\rho_n \leftarrow \sum_{\mathbf{t}} \phi(\mathbf{x}^{(n)}, \mathbf{t}) P_{\theta}(\mathbf{t} | \mathbf{x}^{(n)})$   
     $\mu \leftarrow \mu + \rho_n$   
   $\theta \leftarrow \boldsymbol{\theta}(\mu)$ 
```

- $\phi(\mathbf{x}, \mathbf{t})$: mapping from a labelled example $(\mathbf{x}, \mathbf{t})$ to a vector of sufficient statistics (μ)
- $\boldsymbol{\theta}(\mu)$: maximum likelihood estimate

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   $\mu \leftarrow \mu + \rho_n$ 
   $\theta \leftarrow \boldsymbol{\theta}(\mu)$ 

```

Stepwise EM

```

 $\theta \leftarrow \text{initialization}; \quad k = 0$ 
for each iteration  $w = 1, \dots, W$ :
  for each example  $n = 1, \dots, N$  in random order:
     $\rho_n \leftarrow \sum_{\mathbf{t}} \phi(\mathbf{x}^{(n)}, \mathbf{t}) P_{\theta}(\mathbf{t}|\mathbf{x}^{(n)})$ 
     $\theta \leftarrow \boldsymbol{\theta}((1 - \eta_k) \theta + \eta_k \rho_n)$ 
   $k \leftarrow k + 1$ 

```

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   $\mu \leftarrow \mu + \rho_n$ 
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```

Stepwise EM

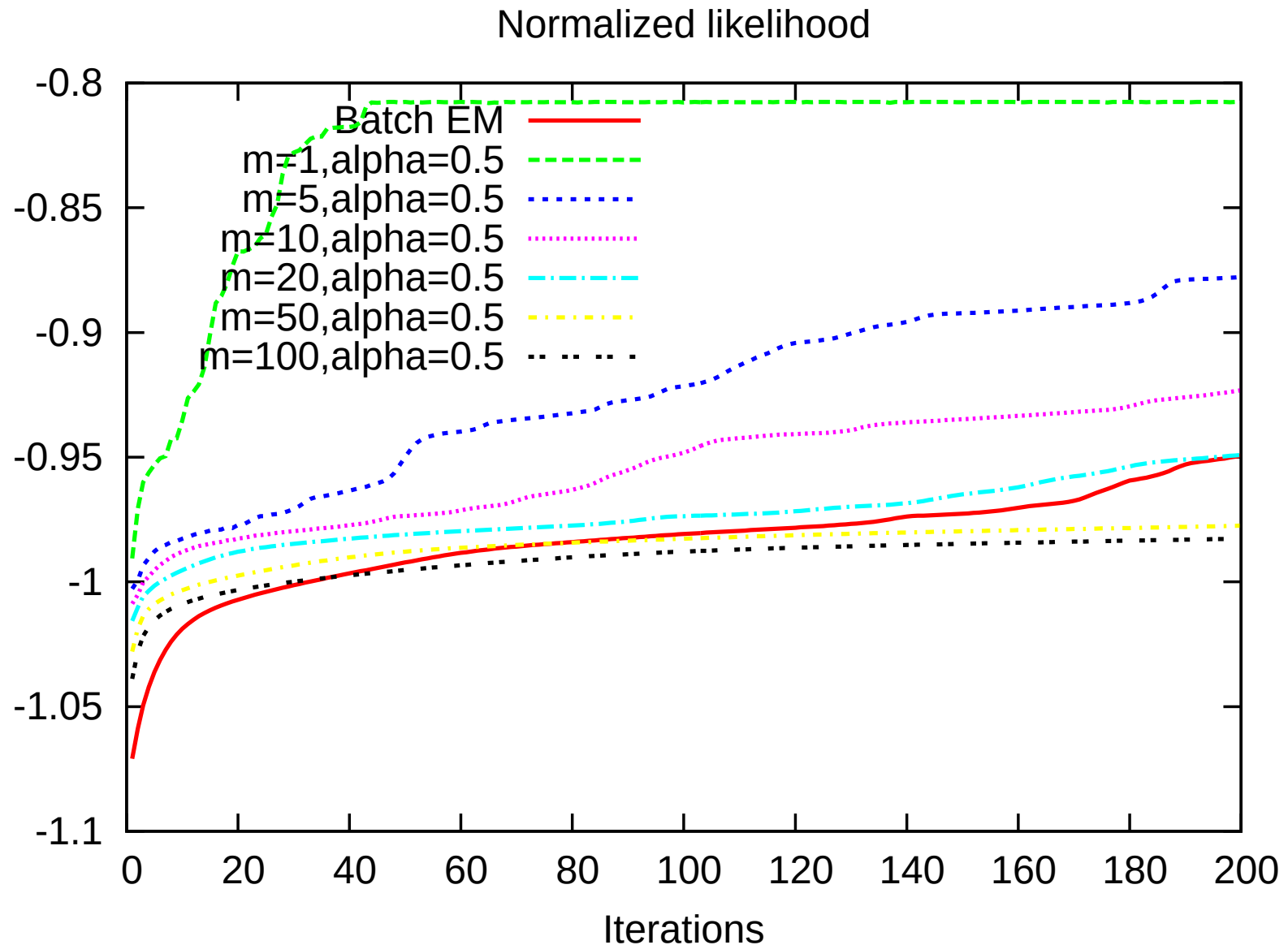
```

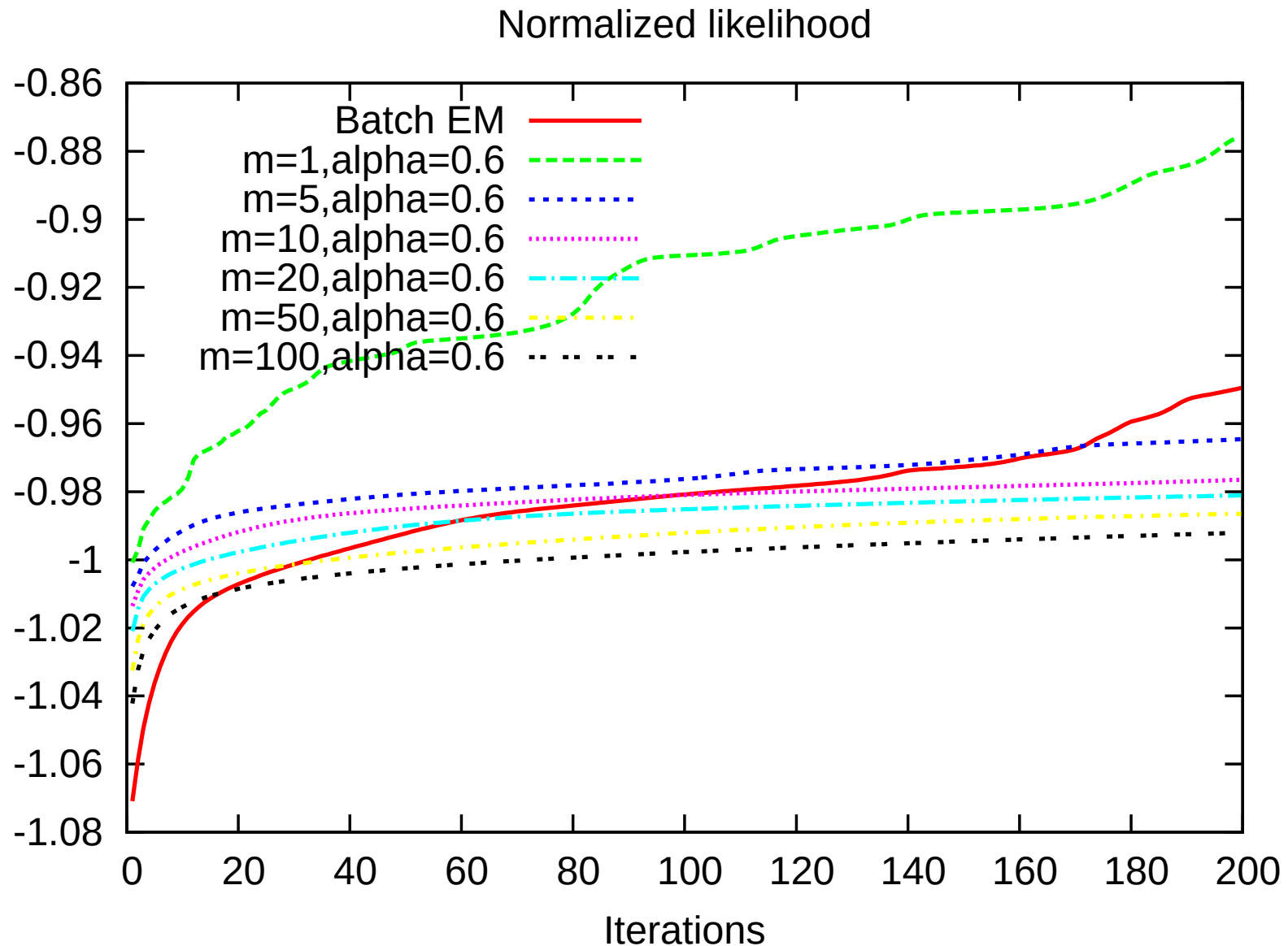
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   $k \leftarrow k + 1$ 

```

- $\phi(\mathbf{x}, \mathbf{t})$: mapping from a labelled example $(\mathbf{x}, \mathbf{t})$ to a vector of sufficient statistics (μ)
- $\boldsymbol{\theta}(\mu)$: maximum likelihood estimate
- Stepwise EM: convergence is guaranteed if $\sum_{k=0}^{\infty} \eta_k = \infty$ and $\sum_{k=0}^{\infty} \eta_k^2 < \infty$
 - $\eta_k = (k + 2)^{-\alpha}$ with $0,5 \leq \alpha \leq 1$
 - Approach: take m examples at once

Palindrome language (15 random initializations, $\alpha = 0,5$)



Palindrome language (15 random initializations, $\alpha = 0,6$)

Palindrome language

[% of palindromes generated in average by the estimated grammar (50K sentences)]

- Batch EM $21,9 \% \pm 10,1$
- Online EM: from random strings

$\alpha \backslash m$	1	5	10	20	50
0,5	$46,7 \pm 20,8$	$35,1 \pm 21,0$	$29,5 \pm 18,3$	$21,8 \pm 10,2$	$16,6 \pm 2,3$
0,6	$34,1 \pm 20,2$	$20,3 \pm 11,3$	$16,8 \pm 2,4$	$15,8 \pm 2,4$	$14,9 \pm 2,8$
0,7	$17,2 \pm 3,1$	$15,1 \pm 2,7$	$14,9 \pm 2,9$	$14,6 \pm 3,3$	$14,9 \pm 3,4$

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0,7	$17,2 \pm 3,1$	$15,1 \pm 2,7$	$14,9 \pm 2,9$	$14,6 \pm 3,3$	$14,9 \pm 3,4$

- Online EM: from short to long strings

$\alpha \backslash m$	1	5	10	20	50
0,5	$82,4 \pm 21,6$	$94,2 \pm 11,6$	$89,6 \pm 19,6$	$92,5 \pm 12,4$	$65,1 \pm 34,9$
0,6	$86,8 \pm 14,2$	$90,8 \pm 13,1$	$92,7 \pm 12,2$	$84,7 \pm 21,7$	$50,7 \pm 29,3$
0,7	$88,8 \pm 14,1$	$81,9 \pm 19,6$	$66,6 \pm 27,1$	$54,5 \pm 27,0$	$33,6 \pm 16,7$

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0,7	$88,8 \pm 14,1$	$81,9 \pm 19,6$	$66,6 \pm 27,1$	$54,5 \pm 27,0$	$33,6 \pm 16,7$

- Online EM: from long to short strings

$\alpha \backslash m$	1	5	10	20	50
0,5	$30,6 \pm 2,0$	$83,4 \pm 11,9$	$84,6 \pm 19,4$	$71,0 \pm 27,7$	$33,2 \pm 26,9$
0,6	$79,6 \pm 13,1$	$70,2 \pm 28,6$	$42,4 \pm 30,6$	$26,0 \pm 18,6$	$16,8 \pm 3,9$
0,7	$52,1 \pm 28,3$	$19,0 \pm 6,5$	$17,0 \pm 4,3$	$16,1 \pm 4,0$	$15,9 \pm 3,7$

Motivation: Supervised learning: (x, y) x : input data (sentence) y : label

- Problem: to annotate data is slow and expensive
- **Active learning: to annotate just the necessary data**

Pool-based active learning algorithm [Settles 2008, Settles 2010]:

```
Given: Labeled set  $\mathcal{L}$ , unlabeled pool  $\mathcal{U}$ ,  
          query strategy  $\phi(\cdot)$ , query batch size  $B$   
repeat  
    // learn a model using the current  $\mathcal{L}$   
     $\theta = \text{train}(\mathcal{L})$   
    for  $b = 1$  to  $B$  do  
        // query the most informative instance  
         $\mathbf{x}_b^* = \arg \max_{\mathbf{x} \in \mathcal{U}} \phi(\mathbf{x})$   
        // move the labeled query from  $\mathcal{U}$  to  $\mathcal{L}$   
         $\mathcal{L} = \mathcal{L} \cup \langle \mathbf{x}_b^*, \text{label}(\mathbf{x}_b^*) \rangle$   
         $\mathcal{U} = \mathcal{U} - \mathbf{x}_b^*$   
    end  
until some stopping criterion
```

➤ **Uncertainty sampling** We select the sample that it has most uncertainty about their label

➤ **Least confidence (LC):**

$$\phi^{LC}(\mathbf{x}) = 1 - P_{\theta}(\hat{\mathbf{t}} | \mathbf{x})$$

➤ **Sequence entropy (SE)**

$$\phi^{SE}(\mathbf{x}) = - \sum_{\hat{\mathbf{t}}} P_{\theta}(\hat{\mathbf{t}} | \mathbf{x}) \log P_{\theta}(\hat{\mathbf{t}} | \mathbf{x})$$

➤ **N -best sequence entropy (NSE)**

$$\phi^{NSE}(\mathbf{x}) = - \sum_{\hat{\mathbf{t}} \in \mathcal{N}} P_{\theta}(\hat{\mathbf{t}} | \mathbf{x}) \log P_{\theta}(\hat{\mathbf{t}} | \mathbf{x})$$

Handwritten Text Recognition experiment [Romero 18]:

- ESPOSALLES dataset: single writer in old Catalan in the 17th century, composed by 173 pages that contain 5,437 lines.

	$\mathcal{L}$	$\mathcal{U}$	Test	Total
Lines	150	4,470	827	5,447
Running words	1,650	50,234	8,893	60,777
Vocabulary	397	2,914	1,119	3,500

- Initial model trained on 150 sentences randomly selected
- Batch size: 150 lines for each selection iteration (about 5 pages)

